

Problem Set 2 — The Protocol

Physics 2026–27 · due at the start of class, Tuesday, September 8

NAME _____ DATE _____

How to work this set. Every quantitative problem gets a labeled **K / D / E / A** — the exact table format from class. A bare final number earns no credit, even if it's correct, and **units travel with the math**: every given gets a unit in K, the target unit gets named in D, and units ride through every line of A. This is the format the **Unit 0 Skills Check (Thursday, September 10)** will require.

1 **Unscramble the protocol** KDEA

A student solved this problem correctly: *"A cart moves at 8.0 m/s. How long does it take to cross a 120 m gym?"* — but their four lines got shuffled. Write **K**, **D**, **E**, or **A** next to each line.

_____ $d = v \, t$

_____ $t = d / v = 120 \, \text{m} \div 8.0 \, \text{m/s} = 15 \, \text{s}$

_____ $v = 8.0 \, \text{m/s}, \quad d = 120 \, \text{m}$

_____ the crossing time t , in seconds

Then answer in one sentence: which *two* letters have to be on the page before E "usually becomes obvious," and why?

2 **The school-zone check** CONVERSION

A car near campus is clocked at **40 km/h**. The posted school-zone limit is **12 m/s**. Run the full protocol to convert the clocked speed to **m/s**, then answer: is the car speeding? (Careful — the raw numbers point one way, the physics may point the other.)

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	EQUATIONS / CONVERSIONS
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

3 The trap, part two

UNITS FIRST

A classmate computes $2 \text{ km} + 350 \text{ m}$ by typing $2 + 350 = 352$.

- In one sentence: why is 352 not just wrong, but *meaningless*?
- Run KDEA and get the real total. (The D step is where you pick the one unit everything must arrive in.)

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	EQUATIONS / CONVERSIONS
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

4 Counting the storm

$D = VT$

Sound travels at 343 m/s . During a summer storm you see a lightning flash and hear the thunder 4.0 s later.

- Run KDEA to find how far away the strike was, in meters.
- Convert your answer to kilometers, and round it to match the precision your inputs actually support.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	EQUATIONS / CONVERSIONS
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

5 Rearrange before you plug

SYMBOLS FIRST

A cart crosses a 2.3 m track in 0.50 s . How fast was it moving? Rearrange $d = vt$ for your unknown **before** a single number goes in — the symbols-first line is part of the grade.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	EQUATIONS / CONVERSIONS
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

6 Your week, in seconds

MULTI-STEP

Block 2 Physics meets three times a week: 44 min Monday, 80 min Tuesday, and 80 min Thursday. Run KDEA to find how many **seconds** of physics you get in one full week. Report the final answer in scientific notation.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	EQUATIONS / CONVERSIONS
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

7 Grade this work

CRITIQUE

The problem: "How long does it take to drive 150 km at 25 m/s?" A student writes, in full:

$$150 / 25 = 6 \text{ hours}$$

- Find at least **two** distinct protocol violations in that line. Be specific — name the KDEA step each one skipped.
- Now do it right: run the full protocol and report the actual time (any correct unit is fine, but say which one).

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	EQUATIONS / CONVERSIONS
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

8 Dimensional detective

UNITS AS EVIDENCE

A speed v is measured in m/s and a time t in s . For each expression below, work out the units it produces (show the cancellation) and say whether it could possibly compute a **distance**.

a. $v \times t$

b. v / t

c. t / v

One of the three produces the units of a quantity we'll meet in Unit 1. If you can guess its name, write it down — no penalty for guessing.

WORK

Graded on the homework 4-3-2-1 scale. The core dogma applies to every problem here: an answer without units does not count — this is science, not math.